VIS 2025

# SimVecVis: A Dataset for Enhancing MLLMs in Visualization Understanding

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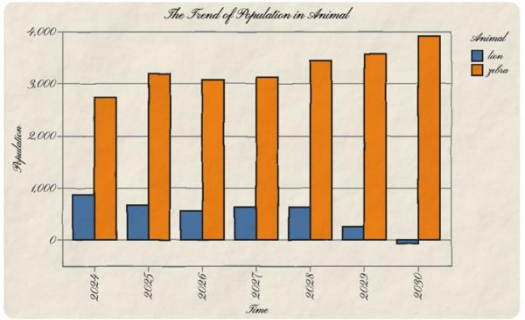
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# MLLMs for Natural Image vs. Visualization Image



what is the cat doing?

It looks like the cat is perched on the edge of an open window or a sliding window frame. The cat seems alert and cautious — possibly trying to explore or find a way down while keeping its balance on the narrow ledge.




what is the value for lion in 2029?

The groundtruth is around 300.

From the chart, the lion population in **2029** is approximately **200**.

MLLMs can precisely understand natural images.

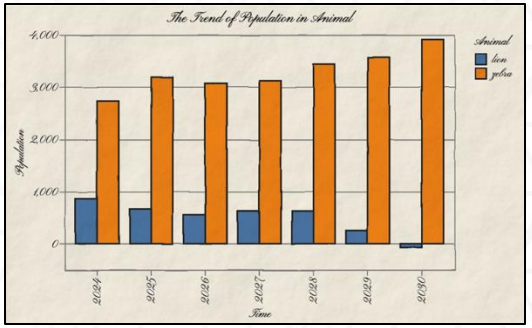
MLLMs struggle to **precisely understand** data visualizations

# Motivation

- Limitation of current MLLMs
  - Originally designed for **natural images**
  - Visualizations = data encoded via rules (axes, legends)
  - Current MLLMs are rarely trained to interpret encoding
- MLLMs struggle to **precisely understand** data visualizations
  - existing representations are either too low-level (pixels) or too structurally heavy (SVG)
  - rare dataset that teaches MLLMs how to reason about data-encoding relationships step-by-step.



Natural Image



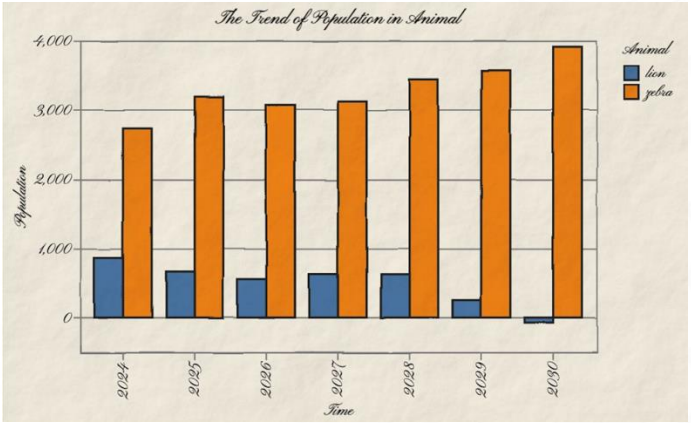

Visualization Image

# SimVecVis Dataset

- Teach models to read charts like humans
  - Understand encodings
  - Locate visual marks
  - Calculate values
- SimVec: represent visualization image using a compact format
  - Locate the position of visual marks
  - Locate the text and identify the text content -> axes, legends, labels

# SimVec: Simplified Vector Format

- Element types: rect, line, polygon, circle, text
- Flatten structure, remove noise
- Standardize coordinates & color (HSL)
- About 90% token reduction vs. SVG

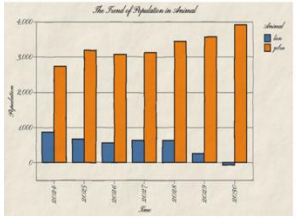



```
text lion [467,44,17,14]|
text zebra [467,57,27,14]|
…
line (0,0,17)
57,29;432,29;432,255;57,255;57,29|
rect (12,8,10) [66,186,20,43]|
rect (2,18,11) [87,92,20,137]|
rect (12,8,10) [118,196,20,33]|
rect (2,18,11) [139,69,20,160]|
rect (12,8,10) [170,201,20,27]|
…
```

# Dataset Construction Pipeline

- Metadata: data attributes, chart type, data content
- Charts in SVG and PNG format
- SimVec
- QA pairs
- CoT sentence

Grouped bar chart
Attributes: C+O+Q



```
text lion
[467,44,17,14]|
text zebra
[467,57,27,14]|
…
line (0,0,17) …
```

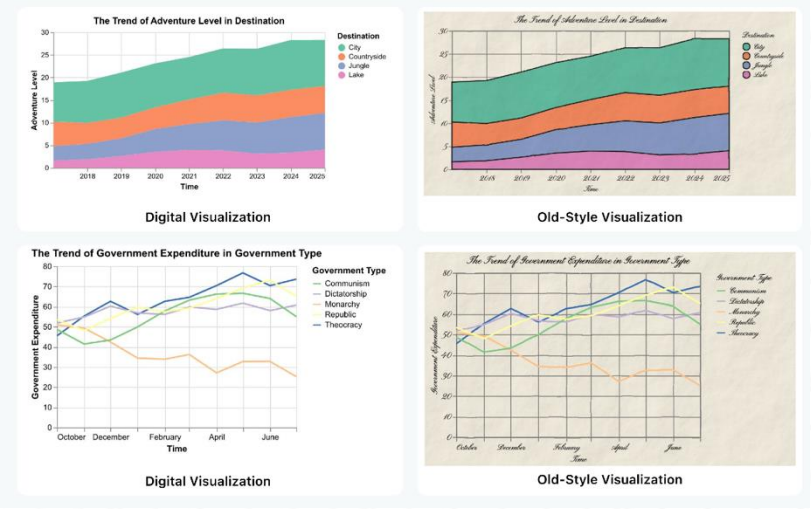
Q: What is the population of 2024 for lion.
A: the population of is 850

The y-axis maps population values from 0 to 400 onto pixel positions from 230 to 29, and the population of lions in 2024 corresponds to pixel position 43.

Metadata | Charts | SimVec | QA pair | CoT sentences

# SimVecVis Dataset Overview

- 3,000 charts
  - 1,012 bar, 1,012 line, 975 area
- 5,642 data QA pairs
  - Includes “historical-style” noisy charts to increase the robustness



# Evaluation

Three Hypotheses for Chart understanding:

- H1: Existing MLLMs cannot accurately support data QA **without** fine-tuning using SimVecVis
- H2: **CoT** reasoning improves data QA accuracy, and is further enhanced with **SimVec** support
- H3: Training MLLMs with SimVec representations improves their ability to reconstruct visualizations

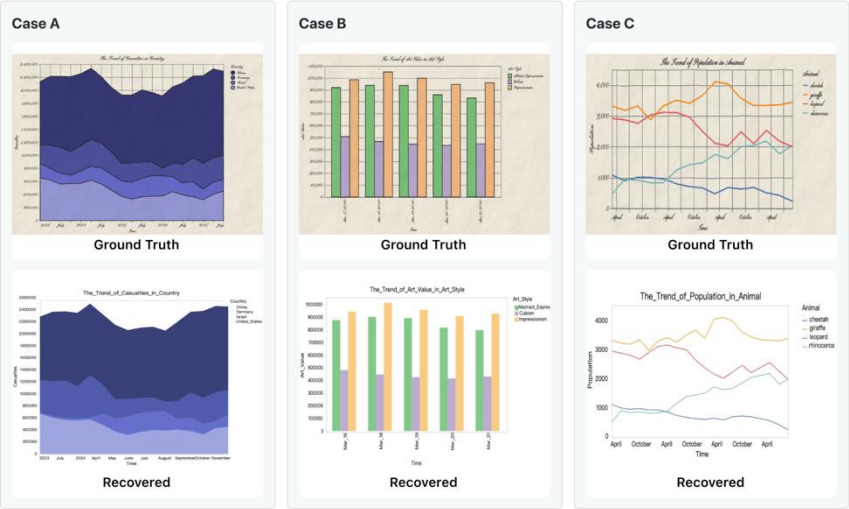
We fine-tune state-of-the-art MiniCPM (7B) to evaluate the effectiveness

# Evaluation- Data QA Accuracy

- Zero-shot performance is low. (H1)
- Adding chain-of-thought improves accuracy. Adding SimVec yields the best performance by a large margin. (H2)

| Model | < 5% | < 10% | < 20% |
|---|---|---|---|
| GPT-4o | 16.54% | 29.62% | 42.69% |
| MiniCPM (zero-shot) | 11.92% | 17.69% | 57.69% |
| DeepSeek-VL(zero-shot) | 10.00% | 17.31% | 26.92% |
| Qwen-VL (zero-shot) | 7.31% | 13.46% | 21.15% |
| MiniCPM (SimVec + QA w/ CoT) | **53.84%** | **69.23%** | **80.77%** |
| MiniCPM (QA w/ CoT) | 29.23% | 45.76% | 69.23% |
| MiniCPM (QA w/o CoT) | 26.92% | 41.92% | 25.38% |
| Qwen-VL (SimVec + QA w/ CoT) | 5.38% | 10.00% | 18.08% |
| Qwen-VL (QA w/ CoT) | 12.31% | 21.54% | 35.77% |
| Qwen-VL (QA w/o CoT) | 11.54% | 19.62% | 31.15% |

# Evaluation - Chart Reconstruction




| Quality Metric | Line | Bar | Area |
|---|---|---|---|
| Text Hit Rate | 99.79% | 99.60% | 99.83% |
| Text Similarity | 98.37% | 96.60% | 98.72% |
| Text Center Distance | 2.89 | 8.26 | 2.70 |
| Element Color Distance | 1.06 | 1.78 | 2.14 |
| Element Position Distance | 8.76 | 10.11 | 29.26 |

*The distance unit is 1/1000 of the image size.

Top: Input bitmap image.
Bottom: reconstructed SimVec.

# SimVecVis: A Dataset for Enhancing MLLMs in Visualization Understanding

- SimVec: compact, consistent representation for visualization
- SimVecVis: dataset with SimVec representation + CoT reasoning
- SimVecVis brings gains in data-centric chart understanding

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github.com/VIDA-Lab/SimVecVis

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